

A Omitted Judgment Rules

A.1 Naturals

$\boxed{\Gamma \vdash a :^\ell A}$ (Type checking)

$$\begin{array}{c} \text{WT-NAT} \\ \vdash \Gamma \\ \hline \Gamma \vdash \text{Nat} :^\ell \mathcal{U}_i \end{array} \quad \begin{array}{c} \text{WT-ZERO} \\ \vdash \Gamma \\ \hline \Gamma \vdash \text{zero} :^\ell \text{Nat} \end{array} \quad \begin{array}{c} \text{WT-SUCC} \\ \Gamma \vdash a :^\ell \text{Nat} \\ \hline \Gamma \vdash \text{succ } a :^\ell \text{Nat} \end{array}$$

$$\begin{array}{c} \text{WT-IND} \\ \Gamma, z :^{\ell_0} \text{Nat} \vdash A :^{\ell_1} \mathcal{U}_i \\ \Gamma \vdash a :^{\ell_0} \text{Nat} \\ \Gamma \vdash b_0 :^{\ell_0} A[\text{zero}/z] \\ \Gamma, x :^{\ell_0} \text{Nat}, y :^{\ell_0} A[x/z] \vdash b_1 :^{\ell_0} A[\text{succ } x/z] \\ \ell_0 \leq \ell \\ \hline \Gamma \vdash \text{ind}^{\ell_0} a b_0 (\lambda x y. b_1) :^\ell A[a/z] \end{array}$$

$\boxed{a \Rightarrow b}$ (Parallel reduction)

$$\begin{array}{c} \text{P-INDZERO} \\ b_0 \Rightarrow b_1 \\ \hline \text{ind}^\ell \text{zero } b_0 (\lambda x y. c_0) \Rightarrow b_1 \end{array} \quad \begin{array}{c} \text{P-INDSUCC} \\ \begin{array}{c} a_0 \Rightarrow a_1 \\ b_0 \Rightarrow b_1 \quad c_0 \Rightarrow c_1 \end{array} \\ \hline \text{ind}^\ell (\text{succ } a_0) b_0 (\lambda x y. c_0) \Rightarrow c_1[a_1/x, \text{ind}^\ell a_1 b_1 (\lambda x y. c_1)/y] \end{array}$$

$\boxed{\Xi \vdash a \equiv^\ell b}$ (Indistinguishability)

$$\begin{array}{c} \text{I-NAT} \\ \hline \Xi \vdash \text{Nat} \equiv^\ell \text{Nat} \end{array} \quad \begin{array}{c} \text{I-ZERO} \\ \hline \Xi \vdash \text{zero} \equiv^\ell \text{zero} \end{array} \quad \begin{array}{c} \text{I-SUCC} \\ \Xi \vdash a_0 \equiv^\ell a_1 \\ \hline \Xi \vdash \text{succ } a_0 \equiv^\ell \text{succ } a_1 \end{array}$$

$$\begin{array}{c} \text{I-IND} \\ \Xi \vdash a_0 \equiv^\ell a_1 \\ \Xi \vdash b_0 \equiv^\ell b_1 \\ \Xi, x : \ell_0, y : \ell_0 \vdash c_0 \equiv^\ell c_1 \\ \ell_0 \leq \ell \\ \hline \Xi \vdash \text{ind}^{\ell_0} a_0 b_0 (\lambda x y. c_0) \equiv^\ell \text{ind}^{\ell_0} a_1 b_1 (\lambda x y. c_1) \end{array}$$

$\boxed{\Xi \vdash a : \ell}$ (Level checking)

$$\begin{array}{c} \text{L-NAT} \\ \hline \Xi \vdash \text{Nat} : \ell \end{array} \quad \begin{array}{c} \text{L-ZERO} \\ \hline \Xi \vdash \text{zero} : \ell \end{array} \quad \begin{array}{c} \text{L-SUCC} \\ \Xi \vdash a : \ell \\ \hline \Xi \vdash \text{succ } a : \ell \end{array} \quad \begin{array}{c} \text{L-IND} \\ \begin{array}{c} \Xi \vdash a : \ell_0 \quad \Xi \vdash b_0 : \ell_0 \\ \Xi, x : \ell_0, y : \ell_0 \vdash b_1 : \ell_0 \\ \ell_0 \leq \ell \end{array} \\ \hline \Xi \vdash \text{ind}^{\ell_0} a b_0 (\lambda x y. b_1) : \ell \end{array}$$

A.2 Context well-formedness

$\boxed{\vdash \Gamma}$ (Context well-formedness)

$$\begin{array}{c} \text{WF-NIL} \\ \vdash \Gamma \\ \hline \vdash \bullet \end{array} \quad \begin{array}{c} \text{WF-CONS} \\ \vdash \Gamma \quad \Gamma \vdash A : \mathcal{U}_i \\ \hline \vdash \Gamma, x :^\ell A \end{array}$$

A.3 Parallel reduction

$$\boxed{a \Rightarrow b}$$

(Parallel reduction)

$$\begin{array}{c} \text{P-VAR} \\ \hline x \Rightarrow x \end{array} \quad \begin{array}{c} \text{P-PI} \\ \hline A_0 \Rightarrow A_1 \quad B_0 \Rightarrow B_1 \\ \hline \Pi x :^{\ell_0} A_0. B_0 \Rightarrow \Pi x :^{\ell_0} A_1. B_1 \end{array} \quad \begin{array}{c} \text{P-ABS} \\ \hline b_0 \Rightarrow b_1 \\ \hline \lambda x^{\ell_0}. b_0 \Rightarrow \lambda x^{\ell_0}. b_1 \end{array} \quad \begin{array}{c} \text{P-APP} \\ \hline b_0 \Rightarrow b_1 \quad a_0 \Rightarrow a_1 \\ \hline b_0 a_0^{\ell_0} \Rightarrow b_1 a_1^{\ell_0} \end{array}$$

$$\begin{array}{c} \text{P-UNIV} \\ \hline \mathcal{U}_i \Rightarrow \mathcal{U}_i \end{array} \quad \begin{array}{c} \text{P-EMPTY} \\ \hline \perp \Rightarrow \perp \end{array} \quad \begin{array}{c} \text{P-ABSURD} \\ \hline b_0 \Rightarrow b_1 \\ \hline \text{absurd } b_0 \Rightarrow \text{absurd } b_1 \end{array} \quad \begin{array}{c} \text{P-EQ} \\ \hline a_0 \Rightarrow a_1 \quad b_0 \Rightarrow b_1 \\ \hline a_0 =^{\ell_0} b_0 \Rightarrow a_1 =^{\ell_0} b_1 \end{array}$$

$$\begin{array}{c} \text{P-REFL} \\ \hline \text{refl} \Rightarrow \text{refl} \end{array} \quad \begin{array}{c} \text{P-J} \\ \hline p_0 \Rightarrow p_1 \quad c_0 \Rightarrow c_1 \\ \hline \mathbf{J} \, c_0 \, p_0^{\ell} \Rightarrow \mathbf{J} \, c_1 \, p_1^{\ell} \end{array} \quad \begin{array}{c} \text{P-SIG} \\ \hline A_0 \Rightarrow A_1 \quad B_0 \Rightarrow B_1 \\ \hline \Sigma x :^{\ell_0} A_0. B_0 \Rightarrow \Sigma x :^{\ell_0} A_1. B_1 \end{array}$$

$$\begin{array}{c} \text{P-PAIR} \\ \hline a_0 \Rightarrow a_1 \quad b_0 \Rightarrow b_1 \\ \hline (a_0^{\ell_0}, b_0) \Rightarrow (a_1^{\ell_0}, b_1) \end{array} \quad \begin{array}{c} \text{P-LET} \\ \hline a_0 \Rightarrow a_1 \quad b_0 \Rightarrow b_1 \\ \hline \text{let } (x^{\ell_0}, y) = a_0^{\ell} \text{ in } b_0 \Rightarrow \text{let } (x^{\ell_0}, y) = a_1^{\ell} \text{ in } b_1 \end{array} \quad \begin{array}{c} \text{P-UNIT} \\ \hline \top \Rightarrow \top \end{array}$$

$$\begin{array}{c} \text{P-TT} \\ \hline \text{tt} \Rightarrow \text{tt} \end{array} \quad \begin{array}{c} \text{P-SEQ} \\ \hline a_0 \Rightarrow a_1 \quad b_0 \Rightarrow b_1 \\ \hline \text{let tt} = a_0^{\ell} \text{ in } b_0 \Rightarrow \text{let tt} = a_1^{\ell} \text{ in } b_1 \end{array} \quad \begin{array}{c} \text{P-NAT} \\ \hline \text{Nat} \Rightarrow \text{Nat} \end{array} \quad \begin{array}{c} \text{P-ZERO} \\ \hline \text{zero} \Rightarrow \text{zero} \end{array}$$

$$\begin{array}{c} \text{P-SUCC} \\ \hline a_0 \Rightarrow a_1 \\ \hline \text{succ } a_0 \Rightarrow \text{succ } a_1 \end{array} \quad \begin{array}{c} \text{P-IND} \\ \hline a_0 \Rightarrow a_1 \\ b_0 \Rightarrow b_1 \quad c_0 \Rightarrow c_1 \\ \hline \text{ind}^{\ell} a_0 \, b_0 \, (\lambda x \, y. c_0) \Rightarrow \text{ind}^{\ell} a_1 \, b_1 \, (\lambda x \, y. c_1) \end{array}$$

$$\boxed{a \Rightarrow^* b}$$

(Multi-step parallel reduction)

$$\begin{array}{c} \text{PS-REFL} \\ \hline a \Rightarrow b \\ \hline a \Rightarrow^* b \end{array} \quad \begin{array}{c} \text{PS-TRANS} \\ \hline a \Rightarrow b \quad b \Rightarrow^* c \\ \hline a \Rightarrow^* c \end{array}$$

A.4 Level checking

$$\boxed{\Xi \vdash a : \ell}$$

(Level checking)

$$\begin{array}{c} \text{L-VAR} \\ \hline x : \ell_0 \in \Xi \quad \ell_0 \leq \ell \\ \hline \Xi \vdash x : \ell \end{array} \quad \begin{array}{c} \text{L-PI} \\ \hline \Xi \vdash A : \ell \\ \Xi, x : \ell_0 \vdash B : \ell \\ \hline \Xi \vdash \Pi x :^{\ell_0} A. B : \ell \end{array} \quad \begin{array}{c} \text{L-ABS} \\ \hline \Xi, x : \ell_0 \vdash b : \ell \\ \hline \Xi \vdash \lambda x^{\ell_0}. b : \ell \end{array} \quad \begin{array}{c} \text{L-APP} \\ \hline \Xi \vdash b : \ell \quad \Xi \vdash a : \ell_0 \\ \hline \Xi \vdash b \, a^{\ell_0} : \ell \end{array}$$

$$\begin{array}{c} \text{L-UNIV} \\ \hline \Xi \vdash \mathcal{U}_i : \ell \end{array} \quad \begin{array}{c} \text{L-EMPTY} \\ \hline \Xi \vdash \perp : \ell \end{array} \quad \begin{array}{c} \text{L-ABSURD} \\ \hline \Xi \vdash \text{absurd } b : \ell \end{array} \quad \begin{array}{c} \text{L-EQ} \\ \hline \Xi \vdash a : \ell_0 \\ \Xi \vdash b : \ell_0 \quad \ell_0 \leq \ell \\ \hline \Xi \vdash a =^{\ell_0} b : \ell \end{array} \quad \begin{array}{c} \text{L-REFL} \\ \hline \Xi \vdash \text{refl} : \ell \end{array}$$

L-J

$$\frac{\Xi \vdash p : \ell_0 \quad \Xi \vdash c : \ell \quad \ell_0 \leq \ell}{\Xi \vdash \mathbf{J} c p^{\ell_0} : \ell}$$

L-SIG

$$\frac{\Xi \vdash A : \ell \quad \Xi, x : \ell_0 \vdash B : \ell}{\Xi \vdash \Sigma x :^{\ell_0} A. B : \ell}$$

L-PAIR

$$\frac{\Xi \vdash a : \ell_0 \quad \Xi \vdash b : \ell}{\Xi \vdash (a^{\ell_0}, b) : \ell}$$

L-LET

$$\frac{\Xi \vdash a : \ell_1 \quad \Xi, x : \ell_0, y : \ell_1 \vdash b : \ell \quad \ell_1 \leq \ell}{\Xi \vdash \mathbf{let} (x^{\ell_0}, y) = a^{\ell_1} \mathbf{in} b : \ell}$$

L-SEQ

$$\frac{\Xi \vdash a : \ell_1 \quad \Xi \vdash b : \ell \quad \ell_1 \leq \ell}{\Xi \vdash \mathbf{let} \mathbf{tt} = a^{\ell_1} \mathbf{in} b : \ell}$$

L-UNIT

$$\frac{}{\Xi \vdash \mathbf{T} : \ell}$$

L-TT

$$\frac{}{\Xi \vdash \mathbf{tt} : \ell}$$

A.5 Normal and Neutral Forms

 $\boxed{\mathbf{ne} a}$

(Neutral forms)

NE-VAR

$$\frac{}{\mathbf{ne} x}$$

NE-APP

$$\frac{\mathbf{ne} b \quad \mathbf{nf} a}{\mathbf{ne} b a^{\ell}}$$

NE-ABSURD

$$\frac{\mathbf{nf} b}{\mathbf{ne} \mathbf{absurd} b}$$

NE-SEQ

$$\frac{\mathbf{ne} a \quad \mathbf{nf} b}{\mathbf{ne} \mathbf{let} \mathbf{tt} = a^{\ell} \mathbf{in} b}$$

NE-J

$$\frac{\mathbf{nf} c \quad \mathbf{ne} p}{\mathbf{ne} \mathbf{J} c p^{\ell}}$$

NE-LET

$$\frac{\mathbf{ne} a \quad \mathbf{nf} b}{\mathbf{ne} \mathbf{let} (x^{\ell_0}, y) = a^{\ell} \mathbf{in} b}$$

 $\boxed{\mathbf{nf} a}$

(Normal forms)

NF-PI

$$\frac{\mathbf{nf} A \quad \mathbf{nf} B}{\mathbf{nf} \Pi x :^{\ell} A. B}$$

NF-ABS

$$\frac{\mathbf{nf} b}{\mathbf{nf} \lambda x^{\ell}. b}$$

NF-UNIV

$$\frac{}{\mathbf{nf} \mathcal{U}_i}$$

NF-VOID

$$\frac{}{\mathbf{nf} \perp}$$

NF-UNIT

$$\frac{}{\mathbf{nf} \mathbf{T}}$$

NF-TT

$$\frac{}{\mathbf{nf} \mathbf{tt}}$$

NF-EQ

$$\frac{\mathbf{nf} a \quad \mathbf{nf} b \quad \mathbf{nf} A}{\mathbf{nf} a =^{\ell} b}$$

NF-REFL

$$\frac{}{\mathbf{nf} \mathbf{refl}}$$

NF-SIG

$$\frac{\mathbf{nf} A \quad \mathbf{nf} B}{\mathbf{nf} \Sigma x :^{\ell} A. B}$$

NF-PAIR

$$\frac{\mathbf{nf} a \quad \mathbf{nf} b}{\mathbf{nf} (a^{\ell}, b)}$$

A.6 Logical Predicate for Unit and Naturals

 $\boxed{\vdash_{\mathbb{N}} a}$

(Natural Values)

N-NE

$$\frac{\mathbf{ne} a}{\vdash_{\mathbb{N}} a}$$

N-ZERO

$$\frac{}{\vdash_{\mathbb{N}} \mathbf{zero}}$$

N-SUCC

$$\frac{\vdash_{\mathbb{N}} a}{\vdash_{\mathbb{N}} \mathbf{succ} a}$$

 $\boxed{[\![\Xi \models_i A]\!] \searrow S}$

(Logical predicate)

SWT-NAT

$$\frac{}{[\![\Xi \models_i \mathbf{Nat}]\!] \searrow \ell. \{a \mid \Xi \vdash a : \ell \wedge \exists b, \vdash_{\mathbb{N}} b \wedge a \Rightarrow^* b\}}$$